

Bootstrap Framework

What is Bootstrap?

Bootstrap is a free, open-source CSS framework originally developed by Twitter. It provides a collection of pre-built CSS classes and JavaScript components that allow developers to build responsive, mobile-first websites quickly and consistently without writing all CSS from scratch.

Bootstrap 5 (used in this course) includes a 12-column grid system, ready-to-use components (buttons, cards, modals, navbars, alerts), and hundreds of utility classes for spacing, colour, and layout.

Bootstrap is added by linking to a CDN in the HTML <head>:

```
<!-- Bootstrap CSS — place in <head> -->
<link rel="stylesheet"
  href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css">

<!-- Bootstrap JS — place before </body> -->
<script
  src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>
```

The Bootstrap Grid System

Bootstrap uses a 12-column responsive grid. Content goes inside .container > .row, then col-* classes define how many of the 12 columns each element occupies at different screen sizes.

```
<div class="container">
  <div class="row">

    <!-- Two equal columns on medium screens and above -->
    <div class="col-12 col-md-6">Left Column</div>
    <div class="col-12 col-md-6">Right Column</div>

    <!-- Three equal columns on large screens -->
    <div class="col-12 col-lg-4">Column 1</div>
    <div class="col-12 col-lg-4">Column 2</div>
    <div class="col-12 col-lg-4">Column 3</div>

  </div>
</div>
```

Class Prefix	Breakpoint	Screen Size
col-	< 576px	Extra small (mobile)
col-sm-	>= 576px	Small (large phone)
col-md-	>= 768px	Medium (tablet)
col-lg-	>= 992px	Large (laptop)
col-xl-	>= 1200px	Extra large (desktop)

Bootstrap Buttons

Bootstrap provides a range of button styles using the btn base class combined with a colour modifier:

```
<button class="btn btn-primary">Primary</button>
<button class="btn btn-secondary">Secondary</button>
<button class="btn btn-success">Success</button>
<button class="btn btn-danger">Danger</button>
<button class="btn btn-outline-primary">Outline</button>
<button class="btn btn-sm btn-warning">Small Warning</button>
<button class="btn btn-lg btn-info">Large Info</button>
```

Bootstrap Alerts

Alerts display contextual feedback messages to the user. They are styled using the alert class and a colour modifier class:

```
<div class="alert alert-success">
  Success! Your form was submitted.
</div>

<div class="alert alert-warning">
  Warning! Please check your input.
</div>

<div class="alert alert-danger">
  Error! Something went wrong.
</div>

<div class="alert alert-info">
  Info: Your session will expire in 5 minutes.
</div>
```

Bootstrap Cards

Cards are flexible content containers with an optional header, body, and footer. They are commonly used for displaying grouped information such as products or profiles.

```
<div class="card" style="max-width: 300px;">
  <div class="card-header bg-primary text-white">
    Card Header
  </div>
  <div class="card-body">
    <h5 class="card-title">Card Title</h5>
    <p class="card-text">Some content for this card.</p>
    <a href="#" class="btn btn-primary btn-sm">Learn More</a>
  </div>
</div>
```

Bootstrap Modal

Modals are dialog boxes that overlay the page. Bootstrap modals require no custom JavaScript they are triggered using data attributes on the button element:

```
<!-- Trigger button -->
<button class="btn btn-primary"
  data-bs-toggle="modal"
  data-bs-target="#myModal">
```

```
Open Modal
</button>

<!-- Modal markup -->
<div class="modal fade" id="myModal" tabindex="-1">
  <div class="modal-dialog modal-dialog-centered">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Modal Title</h5>
        <button class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">
        Modal content goes here.
      </div>
      <div class="modal-footer">
        <button class="btn btn-secondary"
          data-bs-dismiss="modal">Close</button>
      </div>
    </div>
  </div>
</div>
```

Badges & Progress Bars

Badges are small count or label indicators attached to content. Progress bars display a completion percentage.

```
<!-- Badges -->
<span class="badge bg-primary">HTML</span>
<span class="badge bg-success">CSS</span>
<span class="badge bg-warning text-dark">Bootstrap</span>
<span class="badge bg-info text-dark">JavaScript</span>

<!-- Progress Bars -->
<div class="progress mb-2">
  <div class="progress-bar" style="width: 85%">85%</div>
</div>
<div class="progress">
  <div class="progress-bar bg-success" style="width: 80%">80%</div>
</div>
```